

Cost-Aware Generative Portfolios

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Investing through Basis-Portfolios

- Portfolio choice on individual securities is challenging: high-dimensional, estimation uncertainty, unbalanced, idiosyncratic risk, regime-switch, etc.
- Investors allocate capital among portfolios of individual securities, usually characteristics-managed basis-portfolios, for mean-variance efficiency.
 - ME and BE/ME sorts: [Fama and French \(1993\)](#).
 - A Zoo of Factors: [Jensen et al. \(2023\)](#).
 - Correlation-clustered basis assets: [Ahn et al. \(2009\)](#).
 - Characteristic (Efficient) Portfolio: [Daniel et al. \(2020\)](#).
 - Shrinkage (Sparse) Portfolios: [Bryzgalova et al. \(2025\)](#).
- Security-level holdings depend on [Basis-Portfolio Generation](#) and [Basis-Portfolio Allocation](#).

Transaction Cost Reshapes Portfolio Choice

- Transaction costs affect the **joint investment value of characteristics**.
 - Combining characteristics can lower rebalancing costs through offsetting trades: [DeMiguel et al. \(2020\)](#).
- Transaction costs affect **optimal portfolio weights**.
 - Portfolio choice balances expected returns, risk, and rebalancing costs.
 - Cost-aware mean–variance and security-level portfolio choice: [Fan et al. \(2024\)](#); [Jensen et al. \(2026\)](#).
- Transaction costs affect **basis-portfolio generation?**
- Transaction costs affect **the structure of MVE and SDF? More broadly.**

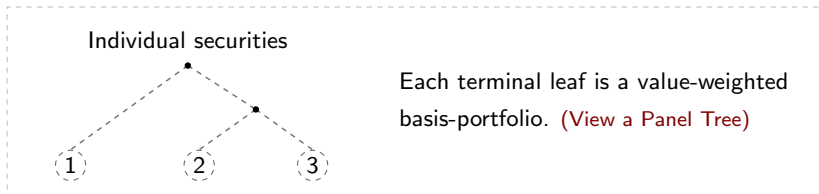
Research Question

*Is **cost-aware allocation** enough to deliver the optimal portfolio choice?
or Transaction Cost reshapes both
the optimal **Basis-Portfolio Generation** and **Basis-Portfolio Allocation**?*

- **Basis-portfolio generation**: which characteristic?
 - Fast-moving characteristics improve gross utility but require frequent security-level rebalancing, e.g. **Short-Term Reversal**.
 - Slow-moving characteristics may offer smaller gross gains with lower transaction costs, e.g. **Quality**.
- Comparison
 - Cost-ignorant basis-portfolio generation + Cost-ignorant allocation
 - Cost-ignorant basis-portfolio generation + Cost-aware allocation
 - Cost-aware basis-portfolio generation + Cost-ignorant allocation
 - Cost-aware basis-portfolio generation + Cost-aware allocation

A Cost-Aware Panel Tree

- We generate basis-portfolios using a **cost-aware Panel Tree**.
 - Panel Tree partitions the cross-section of individual securities into non-overlapping leaf basis-portfolios on lagged characteristics, see [Cong et al. \(2025\)](#).



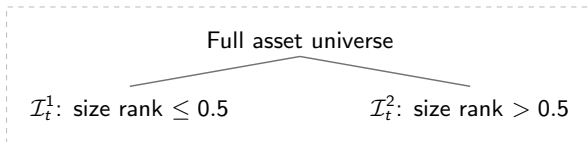
- We select characteristic splits using a cost-aware investment objective, incorporating **transaction costs into basis-portfolio generation**.

Main Findings

- Cost-aware Panel Trees lower turnover and improve **out-of-sample net performance**.
- Most of the net-performance improvement comes from **basis-portfolio generation** rather than subsequent basis-portfolio allocation.
- Cost-aware basis-portfolio generation shifts toward **accounting-fundamental signals**.
 - It retains price-volume signals that remain valuable after costs.
- **Theoretical assumptions and guarantees** that a cost-aware Panel Tree recovers the optimal security-level portfolio choice.

Basis-Portfolio Generation

We partition individual securities using **lagged characteristics**.



Each terminal leaf is a value-weighted basis-portfolio:

$$R_{m,t} = \sum_{i \in \mathcal{I}_t^m} \pi_{i,t} r_{i,t}.$$

- $\pi_{i,t}$: within-leaf value weight based on lagged market equity.
- A security may jump among leaves, as its characteristics evolve.

Stock-Level Holdings and Transaction Costs

Basis-portfolio allocation and within-leaf weights jointly imply holdings:

$$h_{i,t} = w_m \pi_{i,t}, \quad i \in \mathcal{I}_t^m.$$

w_m : basis-portfolio allocation weight.

$\pi_{i,t}$: within-leaf weight.

Following [Brandt et al. \(2009\)](#), we use proportional costs:

$$C_t = \sum_{i \in \mathcal{I}_{t-1} \cup \mathcal{I}_t} \kappa_{i,t} |\Delta h_{i,t}|.$$

- $\Delta h_{i,t}$: trade from pre-trade to target holdings.
- $\kappa_{i,t}$: proportional trading-cost rate varying with **firm size and time**.

Cost-Aware Basis-Portfolio Allocation

We optimize allocation conditional on the generated basis-portfolios.

Mean–variance utility (MVU):

$$\max_w \quad \hat{\mu}_T^\top w - \frac{\delta}{2} w^\top \hat{\Sigma}_T w - \hat{C}_T(w).$$

Maximum Sharpe ratio (SR):

$$\max_w \quad \frac{\hat{\mu}_T^\top w - \hat{C}_T(w)}{\sqrt{w^\top \hat{\Sigma}_T w}}.$$

- $\hat{C}_T(w)$: expected transaction cost.

Growing the Cost-Aware Panel Tree

At each step, we:

1. For each admissible split $\xi = (p, q, m)$, update the leaf basis-portfolios.
2. For each candidate, do basis-portfolio allocation.
3. Select the split with the largest positive **cost-aware gain**.

$$\Delta\mathcal{U}(\xi; \Theta) = \mathcal{U}(\Theta \oplus \xi) - \mathcal{U}(\Theta).$$

- $\mathcal{U}$: optimized **cost-aware objective**.
- Stop when no positive gain remains or the split budget is reached.

Three Theoretical Results

Population mean–variance utility; Theorems 1–2 exclude costs.

- **Theorem 1.**

Characteristic-managed basis-portfolios can approximate optimal security-level investment utility.

- **Theorem 2.**

Panel Tree can recover optimal basis-portfolio utility through recursive characteristic splits.

- **Theorem 3.**

Cost-aware generation can improve net utility beyond reoptimizing allocation on cost-ignorant basis-portfolios.

Interpretation: Panel Tree can recover the investment opportunities available from individual securities. Transaction costs can reshape its basis-portfolio generation, beyond adjustments to allocation.

Data and Empirical Design

Data Sample

- 153 cross-sectionally rank-standardized firm characteristics ([Jensen et al., 2023](#)).
- January 1974 to December 2023

Rolling Window Estimation

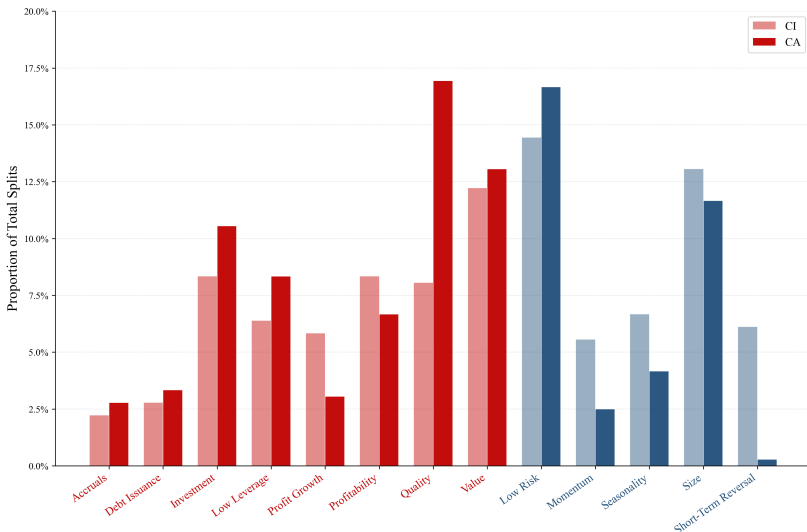
- **Annual** re-estimation using the preceding 10 years.
- **Monthly** updates to firm characteristics and leaf membership.

Cost-ignorant (CI) versus Cost-aware (CA) Panel Trees

- Difference: transaction costs included in the training objective (CA) or excluded (CI).
- Maximum SR and MVU ($\delta = 2$); at most 10 leaves per tree.

Which Characteristic Categories Generate Basis-Portfolios?

Figure 3: Average Split Frequency by Characteristic Category: CA versus CI PTrees



[View heatmap](#)

How Does TC Change Basis-Portfolio Generation?

Table 2: Reallocation of Split Capacity across Rolling Windows

Panel A: Split Shares by Signal Group

Signal group	Frequency share (%)		Score share (%)	
	CI	CA	CI	CA
Accounting-fundamental	54.17	64.72	50.89	61.72
Price-volume	45.83	35.28	49.11	38.28

Panel B: Changes by Characteristic Category

Category (CA minus CI)	Δ Freq.	Δ Score
Quality	+0.80	+4.18
Low risk	+0.20	+1.48
⋮	⋮	⋮
Momentum	−0.28	−1.80
Short-term reversal	−0.53	−1.90

[View example trees](#)

- Split score: weights of 9, 8, ..., 1 in split order.
- Higher frequency and score shares for **accounting-fundamental signals** under CA.

Out-of-Sample Net-of-Cost Performance

Table 3: Out-of-Sample Net-of-Cost Performance

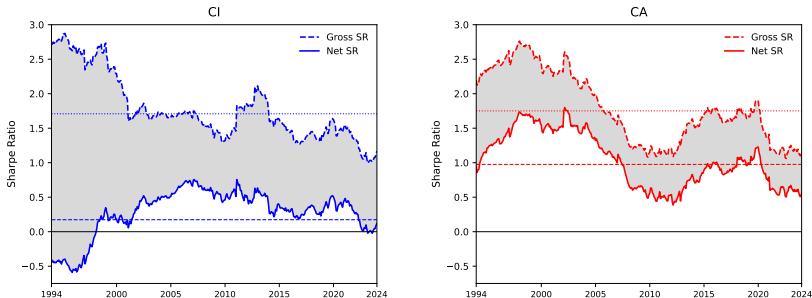
Strategy	Expected Return		Volatility		Objective		TC	Turnover
	Gross	Net	Gross	Net	Gross	Net		
Panel A: MVU								
CI	4.00	1.69	10.34	10.33	2.93	0.62	2.31	351.92
CA	3.89	2.51	10.47	10.58	2.80	1.40	1.38	220.86
Panel B: Maximum SR								
CI	0.63	0.07	1.28	1.32	1.71	0.17	0.56	85.87
CA	0.80	0.46	1.57	1.63	1.75	0.98	0.34	53.16

Expected Return, volatility, TC and turnover: monthly percentages.

- Lower turnover and transaction costs under CA.
- Net performance gains mainly come from higher net returns.

Are Net Performance Gains Persistent over Time?

Figure 5: Rolling Sharpe Ratio Curve Comparison



- CA has lower gap (transaction cost).
- CA net-utility is higher over most of the evaluation period.

Performance beyond Standard Factor Models

Table 5: Factor-Model Regressions for Panel Tree Returns

Factor model	Maximum SR				MVU			
	CI Gross	CA Gross	CI Net	CA Net	CI Gross	CA Gross	CI Net	CA Net
CAPM	0.60*** (9.31)	0.77*** (8.71)	0.04 (0.62)	0.43*** (5.01)	3.69*** (7.46)	3.44*** (7.70)	1.38*** (2.89)	2.07*** (4.59)
FF3	0.58*** (9.55)	0.74*** (8.67)	0.02 (0.26)	0.41*** (4.87)	3.61*** (7.47)	3.51*** (7.44)	1.31*** (2.77)	2.15*** (4.51)
FF5	0.48*** (8.32)	0.65*** (7.91)	-0.08 (-1.37)	0.31*** (3.87)	2.91*** (6.05)	3.37*** (6.15)	0.60 (1.27)	1.98*** (3.59)
FF5 Augmented	0.45*** (7.72)	0.64*** (7.60)	-0.12* (-1.95)	0.29*** (3.61)	2.47*** (5.05)	3.08*** (5.46)	0.17 (0.35)	1.69*** (2.98)
Q5	0.39*** (5.98)	0.60*** (6.99)	-0.16** (-2.31)	0.26*** (3.14)	2.33*** (4.76)	2.53*** (4.35)	0.12 (0.25)	1.17** (2.02)
ALL	0.39*** (6.43)	0.60*** (6.64)	-0.15** (-2.40)	0.26*** (3.02)	2.02*** (4.19)	2.65*** (4.74)	-0.17 (-0.36)	1.30** (2.35)

Monthly alpha (%); Newey–West t in parentheses.

- CA retains positive and significant net alphas across all six factor models under both objectives.

Generation versus Allocation

Basis-portfolio generation	Basis-portfolio allocation	
	Cost-ignorant	Cost-aware
Cost-ignorant	CI+CI	CI+CA
Cost-aware	CA+CI	CA+CA

- CI+CA: retain CI generation and use CA allocation.
- CA+CI: use CA generation, then apply the CI allocation rule.

Where Do the Gains Come From?

Table 4: Basis-Portfolio Generation versus Allocation

Strategy	Maximum SR		MVU	
	Net SR	Gain recovered	Net MVU	Gain recovered
CI–CI	0.17	–	0.62	–
CI–CA	0.43	32.26%	0.83	26.13%
CA–CI	0.91	91.77%	1.40	100.87%
CA–CA	0.98	100.00%	1.40	100.00%

Cost-aware generation recovers **nearly all** of the gain!

Summary

- Question

Is cost-aware allocation enough?

Or should transaction costs also enter **basis-portfolio generation**?

- A Cost-Aware Panel Tree

- Transaction costs enter the criterion for selecting characteristic splits.
- Basis-portfolios are generated to improve net investment performance.

- Empirical Findings

- CA trees deliver lower turnover and higher net performance than CI trees.
- Cost-aware generation recovers **nearly all** of the gain.
- Greater use of **accounting-fundamental signals** under CA.

References I

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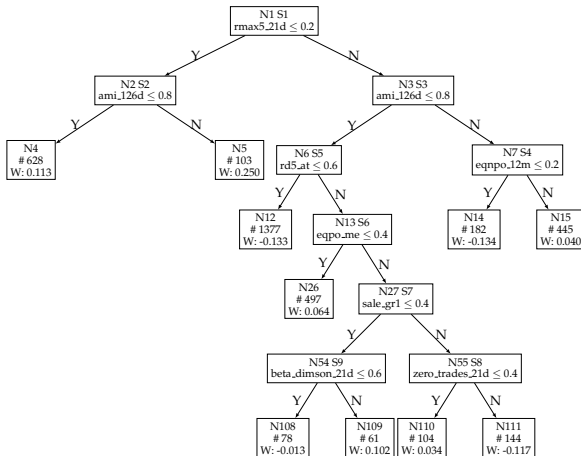
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Cost-Ignorant Panel Tree

Maximum-SR objective. Training window: January 2013–December 2022.

Figure 1: Cost-Ignorant Panel Tree Diagram



[View cost-aware tree](#)

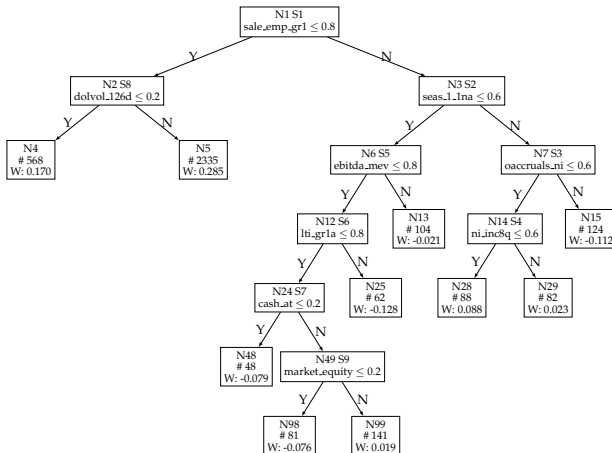
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Cost-Aware Panel Tree

Maximum-SR objective. Training window: January 2013–December 2022.

Figure 2: Cost-Aware Panel Tree Diagram



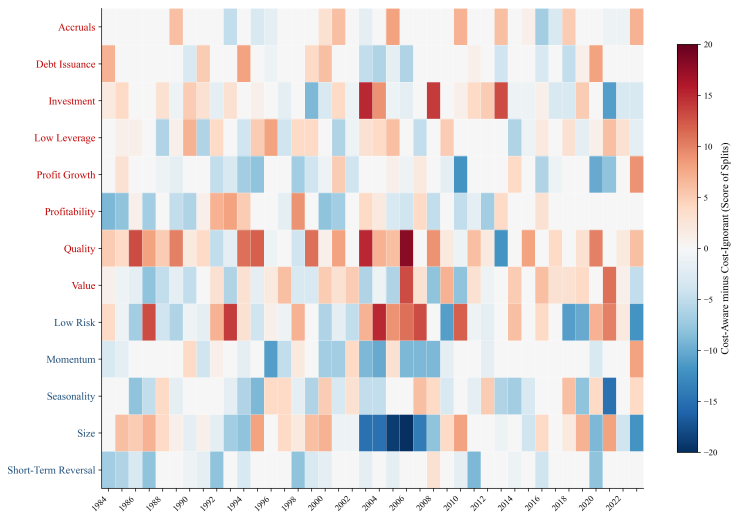
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How Does CA Panel Trees Change Split Importance?

Figure 4: Difference in Score-weighted Split Importance: CA minus CI



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